

Instructions for Students on Taking 2026 USA-NA-AIO Round 1

Contest Date and Time

1. The competition will take place on **January 30, 2026, from 12:00pm to 3:00pm (U.S. Eastern Standard Time)**. All contestants must take the competition within this same time window, regardless of their location.
2. Contest problems will appear in your **student** (not proctor) Edvistas portal **only during the contest window**.
3. Before the competition starts, you will receive a **4-digit Digital Competition Passcode** from your proctor. This passcode is unique for each test site. You must enter this code during the contest window to access the contest problems.
4. You must begin the competition **within 15 minutes** of the official start time. Otherwise, you will not be allowed to take the competition.

Contest Problems

1. There are **9 problems**. **The total score is 300**.
2. Each problem may include multiple parts. Some problems/parts are coding tasks and some are non-coding (math or logical) tasks.
3. There are some multiple choice questions. In each multiple choice question, there is exactly one choice that is correct. If your answer is correct, you get full points. If your answer is wrong or you leave it blank, you get 0 point.
4. Each coding task may have certain restrictions, such as prohibiting the import of additional libraries, modules, or methods beyond those provided. Do not attempt to violate these rules. Doing so will result in a significant penalty, possibly as severe as receiving **0 points**.

Submitting Solutions

1. There are **9 problems**. You must put your solution for each problem in a **separate .ipynb file** using the template provided by USAAIO. Here is the URL:
https://drive.google.com/drive/folders/1rLMBL8TGMUjfytfYbDwgVrR0GEzwoqn6?usp=drive_link
2. Do **not** directly edit the provided template file, as your edits will not be saved. For each problem, make a copy of the template and name it using the following format:

LastName_FirstName_School_ProblemNumber.ipynb

You must strictly follow this naming rule; otherwise, USAAIO may lose track of your submission.

3. You must submit all solutions using the Google Form provided by USAAIO. Here is the URL:
<https://forms.gle/h6nSkdwE7ofbABkx6>
4. You may submit the form **only once**, and you must submit it **before 3:00pm EST on January 30, 2026**. The form will be disabled after the contest ends.
5. If you do any scratch work on paper, write your name on each page and leave them with your proctor. You are not allowed to take them with you after the competition.

Allowed Tools During the Competition

You must have access to the following:

1. Your student Edvistas portal (for reading contest problems).
2. USAAIO's Google Form (for submitting solutions).
3. USAAIO's ipynb template files (to create your solution files).
4. Your registered email (for receiving real-time announcements and asking questions).
5. USAAIO's official website (to check contest rules and permitted URLs).
6. Recording software such as OBS (for screen and face recording).
7. Your registered Google-account email (not your school email), which you must use to access the Google Form and template files.
8. Blank paper and pens/pencils for scratch work.
9. Google Colab's autocompletion feature.



```
1 np.linalg.svd()
```

```
(a: _ArrayLikeInt_co, full_matrices: bool = ...,  
compute_uv: Literal[True] = ..., hermitian: bool =  
...) -> SVDResult
```

Singular Value Decomposition.

When `a` is a 2D array, and `full_matrices=False`, then it is factorized as `u @ np.diag(s) @ vh = (u * s) @ vh`, where `u` and the Hermitian transpose of `vh` are 2D arrays with orthonormal columns and `s` is a 1D array of `a`'s singular values. When `a` is higher-dimensional, SVD is applied in stacked mode as explained below.

1/5

10. Google Colab's help() command.

```
1 import numpy as np  
2  
3 help(np.random.randint)
```

... Help on method randint in module numpy.random.mtrand:

randint(low, high=None, size=None, dtype=<class 'int'>) method of numpy.random.mtrand.RandomState instance
randint(low, high=None, size=None, dtype=int)

Return random integers from 'low' (inclusive) to 'high' (exclusive).

Return random integers from the "discrete uniform" distribution of the specified dtype in the "half-open" interval ['low', 'high'). If 'high' is None (the default), then results are from [0, 'low').

.. note::

New code should use the `~numpy.random.Generator.integers` method of a `~numpy.random.Generator` instance instead; please see the :ref:'random-quick-start'.

Parameters

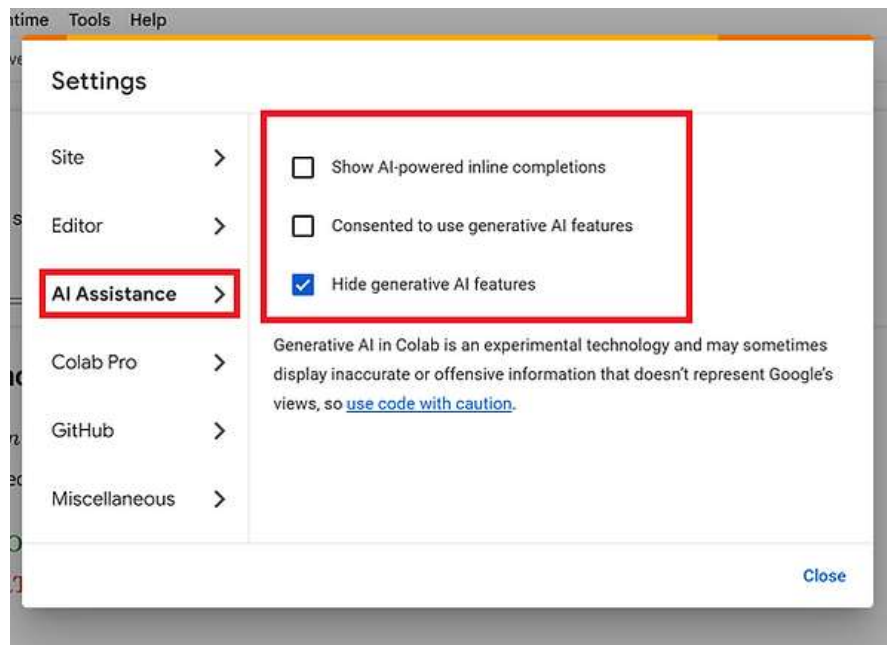
low : int or array-like of ints

Lowest (signed) integers to be drawn from the distribution (unless

Prohibited Behaviors During the Competition

The following are strictly prohibited:

1. Bringing any cheat sheet (paper or digital).
2. Bringing or using a calculator.
 - You may use code cells in Colab for calculations.
3. Using any AI tools, including but not limited to ChatGPT or Google Gemini.
4. Google Colab's integrated AI features must be disabled.



Using Google Colab's integrated AI tools, including but not limited to using light-grey AI-generated code to hint you to write your own code and using the AI tool to debug, is strictly prohibited.

5. Accessing unauthorized apps or websites, including:
 - (1) Search engines (Google, Bing, etc.)
 - (2) Social media (Discord, forums, WhatsApp, WeChat, Xiaohongshu)
 - (3) Past or sample contest problems
 - (4) Non-Google Colab IDEs or debugging tools
 - (5) Having a cell phone during the competition or bringing it to the restroom.
 - (6) Taking screenshots or photos of contest problems.
6. Contest problems are read-only inside the Edvistas portal. You may not capture them in any form.

Non-Coding (Math) Tasks

1. Some tasks require you to write math in text cells. You must typeset all text and math.
2. It is unacceptable to handwrite equations on paper, take a photo, and paste it into your ipynb file.

Self-Recording Requirements

During the contest, you should record:

1. Your computer screen(s).
2. Your face.

3. OBS is free software tool to support your recording.

Additional rules:

3. Install your recording software beforehand and ensure it works.
4. Start recording **before** opening contest problems that will appear in your Edvistas portal.
5. Stop recording **after** submitting your Google Form for your solutions.
6. Please submit the URLs of your recordings with the following Google Form (different from the solution submission form):
<https://forms.gle/TnYNvtNTvKP1bUWo7>
7. You may submit that form **only once**.
8. This form is open **from January 30, 2026, 5pm EST to January 31, 2026, 11:59pm EST**.
9. Your recording links must remain valid until **April 30, 2026**.
10. There is no need to record with a high resolution. USAAIO only requires a reasonable resolution that can be used for the academic honesty check.
11. If your recording file is too big, you may split it into multiple smaller clips and submit URLs for all those clips.

Asking Questions During the Competition

1. Email us at **contact@usaaio.org**.
2. We will try our best to respond, but we may refuse unreasonable questions such as asking how to solve a contest problem.

USAAIO's Announcements During the Competition

1. USAAIO may send out announcements during the competition, such as clarifying anything or correcting any mistake (if any).
2. Announcements will be in the email format that will be sent to the email that you will use during the competition.

Examples of Suspicious Behaviors

The following behaviors will be treated as suspicious:

1. Leaving the computer for a long time (e.g., 20 minutes) and then suddenly solving many problems extremely quickly.
2. Frequently leaving the computer throughout the contest.

3. Showing no visible typing or writing, yet having text, equations, or code appear on the screen as if auto-generated.
4. Starting the recording after seeing contest problems.
5. Pausing the recording at any time.
6. Blocking the webcam or your face.
7. Stopping the recording before submitting your Google Form.
8. Opening social networking apps during the contest (Discord, forums, WhatsApp, WeChat, Xiaohongshu).
9. **Bottom line:**
It is impossible to list all cheating methods. Any attempt to cheat—even if not listed—will result in disqualification.

Disqualification (Zero-Tolerance Policy)

1. **One-cheating, permanent-ban policy:**
If a student is found to have cheated, s/he will be permanently disqualified from the current and **all future** competition cycles.
2. USAAIO will notify the student's proctor.
3. If an educational program (college, summer program, scholarship, etc.) requests information about the student's USAAIO history, we will truthfully disclose the cheating incident.

Discussing Contest Problems After the Competition

1. If you finish early, you may not discuss contest problems with anyone until the contest ends.
2. You may not discuss contest problems online until **February 9, 2026, 11:59pm EST**, unless USAAIO releases the problems earlier.